

DBMS Experiment 2 - Classroom Allotment System

Title

DDL Commands with Constraints – PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK

AIM

To add PRIMARY KEY, FOREIGN KEY, UNIQUE and CHECK constraints to the Classroom Allotment System tables using DDL commands.

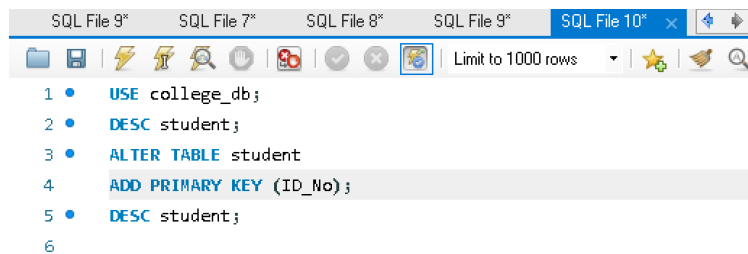
Questions with SQL Code

1) Alter STUDENT table - Add PRIMARY KEY

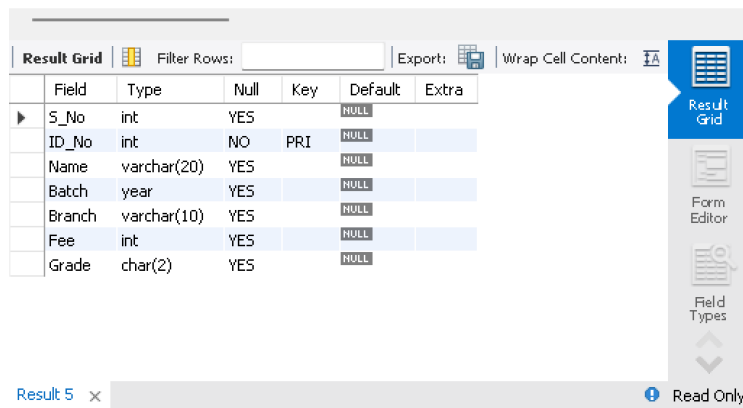
```
ALTER TABLE student
```

```
ADD PRIMARY KEY (ID_No);
```

```
DESC student;
```



```
SQL File 9*  SQL File 7*  SQL File 8*  SQL File 9*  SQL File 10* x
Limit to 1000 rows
1 • USE college_db;
2 • DESC student;
3 • ALTER TABLE student
4 • ADD PRIMARY KEY (ID_No);
5 • DESC student;
6
```



	Field	Type	Null	Key	Default	Extra
►	S_No	int	YES		NULL	
	ID_No	int	NO	PRI	NULL	
	Name	varchar(20)	YES		NULL	
	Batch	year	YES		NULL	
	Branch	varchar(10)	YES		NULL	
	Fee	int	YES		NULL	
	Grade	char(2)	YES		NULL	

Result 5 x Read Only

2) Alter FACULTY table - Add PRIMARY KEY and CHECK

```
ALTER TABLE faculty  
ADD PRIMARY KEY (FacultyID);
```

```
ALTER TABLE faculty  
ADD CONSTRAINT chk_students  
CHECK (NoOfStudents > 0);
```

DESC faculty;

The screenshot shows a SQL IDE with multiple tabs. The active tab, 'SQL File 10*', contains the following SQL queries:

```
1 • ALTER TABLE faculty  
2   ADD PRIMARY KEY (FacultyID);  
3  
4 • ALTER TABLE faculty  
5   ADD CONSTRAINT chk_students  
6   CHECK (NoOfStudents > 0);  
7  
8 • DESC classroom_allotment;
```

Below the queries, the 'Result Grid' displays the structure of the 'faculty' table:

	Field	Type	Null	Key	Default	Extra
▶	S_No	int	YES		NULL	
	Fe S_No ame	varchar(30)	YES		NULL	
	Faculty_ID	varchar(10)	YES		NULL	
	Subject	varchar(30)	YES		NULL	
	Classroom	varchar(15)	YES		NULL	
	No_Of_Students	int	YES		NULL	

On the right side of the Result Grid, there are icons for 'Result Grid', 'Form Editor', and 'Field Types'. The status bar at the bottom indicates 'Result 3' and 'Read Only'.

3) Alter DEPARTMENT table - Add PRIMARY KEY

```
ALTER TABLE department  
ADD PRIMARY KEY (DeptNo);
```

DESC department;

SQL File 8* SQL File 9* SQL File 10* SQL File 10* SQL File 11* x

Limit to 1000 rows

```

1 • USE college_db;
2
3 • DESC department;
4
5 • ALTER TABLE department
6   ADD PRIMARY KEY (DeptNo);
7
8 • DESC department;

```

Result Grid

Filter Rows: Export: Wrap Cell Content: TA

	Field	Type	Null	Key	Default	Extra
▶	DeptNo	int	NO	PRI	NULL	
	DeptName	varchar(20)	YES		NULL	
	DeptHead	varchar(30)	YES		NULL	

Result Grid

Form Editor

Field Types

Result 2 x Read Only

4) Alter COURSE table - Add PRIMARY KEY and CHECK

ALTER TABLE course

ADD PRIMARY KEY (CourseNo);

ALTER TABLE course

ADD CONSTRAINT chk_sem

CHECK (SemNo BETWEEN 1 AND 8);

DESC course;

SQL File 9* SQL File 10* SQL File 10* SQL File 11* SQL File 12*

Limit to 1000 rows

```

1 • USE college_db;
2
3 • ALTER TABLE course
4   ADD PRIMARY KEY (CourseNo);
5
6 • ALTER TABLE course
7   ADD CONSTRAINT chk_sem
8   CHECK (SemNo BETWEEN 1 AND 8);
9
10 • DESC course;

```

Result Grid

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(5)	NO	PRI	NULL	
CourseName	varchar(25)	YES		NULL	
NoOfStudents	int	YES		NULL	
Classroom	varchar(10)	YES		NULL	
FacultyID	varchar(10)	YES		NULL	
Branch	varchar(10)	YES		NULL	
Credits	int	YES		NULL	

Export: | Wrap Cell Content: |

Result Grid

Form Editor

Field Types

5) Alter STUDENT table - Add UNIQUE KEY

ALTER TABLE student

ADD CONSTRAINT uq_student_name UNIQUE(Name);

DESC student;

SQL File 10^o SQL File 10^o SQL File 11^o SQL File 12^o SQL File 13^o

Limit to 1000 rows

```

1 • USE college_db;
2 • ALTER TABLE student
3   ADD CONSTRAINT uq_name UNIQUE (Name);
4 • DESC student;

```

Result Grid

Field	Type	Null	Key	Default	Extra
S_No	int	YES		NULL	
ID_No	int	NO	PRI	NULL	
Name	varchar(20)	YES	UNI	NULL	
Batch	year	YES		NULL	
Branch	varchar(10)	YES		NULL	
Fee	int	YES		NULL	
Grade	char(2)	YES		NULL	

Result Grid
Form Editor
Field Types

6) Add FOREIGN KEY between COURSE and FACULTY

ALTER TABLE course

ADD CONSTRAINT fk_course_faculty

FOREIGN KEY (FacNo)

REFERENCES faculty(FacultyID);

DESC course;

SQL File 10* SQL File 11* SQL File 12* SQL File 13* SQL File 14*

Limit to 1000 rows

```

1 • ALTER TABLE course
2   ADD CONSTRAINT fk_course_faculty
3   FOREIGN KEY (FacNo)
4   REFERENCES faculty(FacultyID);
5 • DESC course;
6

```

Result Grid Filter Rows: Export: Wrap Cell Content: Result Grid Form Editor Field Types

	Field	Type	Null	Key	Default	Extra
▶	CourseNo	varchar(5)	NO	PRI	NULL	
	CourseName	varchar(25)	YES		NULL	
	NoOfStudents	int	YES		NULL	
	Classroom	varchar(10)	YES		NULL	
	FacultyID	varchar(10)	YES		NULL	
	Branch	varchar(10)	YES		NULL	
	Credits	int	YES		NULL	

Result

PRIMARY KEY, FOREIGN KEY, UNIQUE and CHECK constraints were successfully added and verified for the Classroom Allotment System database.